

Lecture 12

Partial derivatives

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For a function $f = f(x, y)$ of two variables, we define the partial derivative of f with respect to x as

$$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h},$$

and similarly for the partial derivative of f with respect to y . To take a partial derivative with respect to a variable, take the derivative with respect to that variable treating all other variables as constants. As an example, consider

$$f(x, y) = 2x^3y^2 + y^3.$$

We have

$$\frac{\partial f}{\partial x} = 6x^2y^2, \quad \frac{\partial f}{\partial y} = 4x^3y + 3y^2.$$

Second derivatives are defined as the derivatives of the first derivatives, so we have

$$\frac{\partial^2 f}{\partial x^2} = 12xy^2, \quad \frac{\partial^2 f}{\partial y^2} = 4x^3 + 6y;$$

and for continuous differentiable functions, the mixed second partial derivatives are independent of the order in which the derivatives are taken,

$$\frac{\partial^2 f}{\partial x \partial y} = 12x^2y = \frac{\partial^2 f}{\partial y \partial x}.$$

To simplify notation, we introduce the standard subscript notation for partial derivatives,

$$f_x = \frac{\partial f}{\partial x}, \quad f_y = \frac{\partial f}{\partial y}, \quad f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad \text{etc.}$$

The Taylor series of $f(x, y)$ about the origin is developed by expanding the function in a multivariable power series that agrees with the function value and all its partial derivatives at $(x, y) = (0, 0)$. We have

$$f(x, y) = f + f_x x + f_y y + \frac{1}{2!} (f_{xx} x^2 + 2f_{xy} xy + f_{yy} y^2) + \dots$$

The function and all its partial derivatives on the right-hand side are evaluated at $(0, 0)$ and are constants. By taking partial derivatives it is evident that $f(0, 0) = f$, $f_x(0, 0) = f_x$, $f_y(0, 0) = f_y$, and so on as the infinite series continues.

Problems for Lecture 12

1. Compute the three partial derivatives of

$$f(x, y, z) = \frac{1}{(x^2 + y^2 + z^2)^n}.$$

2. Given the function $f = f(t, x)$, find the Taylor series expansion of the expression

$$f(t + \alpha \Delta t, x + \beta \Delta t f(t, x))$$

to first order in Δt .

Solutions to the Problems